

Module 02: Agents in Middle School

Learning Objectives

1. Define **Agents** within the context of Middle School.
2. Analyze how Agents interacts with other components of the Active Inference framework.
3. Apply specific constraints of Middle School to the formal definition of Agents.

Introduction

This module explores **Agents**. In the **Middle School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Agents is a critical component of the 8-part Active Inference spine, bridging the gap between Systems and Perception.

Key Concepts

1. Agents as a Markov Blanket Boundary

How does Agents define the boundary between the agent and the environment?

2. Generative Models of Agents

What parameters involved in Agents must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Agents drive the perception-action loop?

Applications

In Middle School, we see Agents manifest in: * **Specific Example 1:** In Scratch or Python, you can build a simple chatbot agent that reads what a user types (perception), looks up a

response in its list of known answers (cognition), and replies with text on the screen (action). Even this basic program has the core structure of an agent -- it senses input, processes it against an internal model, and takes action. * **Specific Example 2:** Enemy characters in video games are agents with simple generative models. A ghost in Pac-Man has sensors (it knows where Pac-Man is), a goal (catch or flee), and movement rules. Each ghost uses a slightly different strategy -- one chases directly, one tries to ambush, one acts randomly. These different internal models produce different behaviors from agents that all live in the same game world.

Conclusion

Understanding Agents allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.